

1 Elastic Launch Energy Worksheet

2 Definitions

Elastic potential energy (stored in a stretched rubber band or spring) is:

$$E_{\text{elastic}} = \frac{1}{2}kx^2 \quad (1)$$

Kinetic energy (KE) is the energy of motion:

$$E_k = \frac{1}{2}mv^2 \quad (2)$$

At a launch angle of 45° , the horizontal distance traveled is:

$$D = \frac{v^2}{g} \quad (3)$$

3 Example Problems

Example 1:

$$E_{\text{elastic}} = \frac{1}{2}(20)(0.15)^2 = 0.225 \text{ J}$$

Example 2:

$$E_k = \frac{1}{2}(0.05)(6)^2 = 0.9 \text{ J}$$

Example 3:

$$D = \frac{8^2}{9.8} = 6.53 \text{ m}$$

4 Exercises

4.1 Elastic Energy

1.

$$E_{\text{elastic}} = \frac{1}{2}(10)(0.2)^2 = 0.20 \text{ J}$$

2.

$$E_{\text{elastic}} = \frac{1}{2}(25)(0.1)^2 = 0.125 \text{ J}$$

3.

$$E_{\text{elastic}} = \frac{1}{2}(30)(0.05)^2 = 0.0375 \text{ J}$$

4.2 Convert Elastic Energy to KE

Assume no energy loss ($E_{\text{elastic}} = E_k$)

1.

$$E_k = \frac{1}{2}(15)(0.2)^2 = 0.30 \text{ J}$$

2. For a 4: find velocity

$$v = \sqrt{\frac{2E_k}{m}} = 2.45 \text{ m/s}$$

3.

$$E_k = \frac{1}{2}(40)(0.1)^2 = 0.20 \text{ J}$$

4. For a 4: find velocity

$$v = \sqrt{\frac{2E_k}{m}} = 2.83 \text{ m/s}$$

4.3 Range at 45 Degrees

1.

$$D = \frac{5^2}{9.8} = 2.55 \text{ m}$$

2.

$$D = \frac{7^2}{9.8} = 5.0 \text{ m}$$

3.

$$D = \frac{10^2}{9.8} = 10.2 \text{ m}$$

4.

$$D = \frac{9^2}{9.8} = 8.27 \text{ m}$$

5. For a 4: solve for v

$$v = \sqrt{Dg} = 7.67 \text{ m/s}$$

6. For a 4: solve for v

$$v = \sqrt{Dg} = 8.86 \text{ m/s}$$